EXAMPLE 4.27

A capacitor of capacitance $\frac{10^{-4}}{\pi}F$, an inductor of inductance $\frac{2}{\pi}H$ and a resistor of resistance 100 Ω are connected to form a series *RLC* circuit. When an AC supply of 220 V, 50 Hz is applied to the circuit, determine (i) the impedance of the circuit (ii) the peak value of current flowing in the circuit (iii) the power factor of the circuit and (iv) the power factor of the circuit at resonance.

Solution

$$L=\frac{2}{\pi}H; C=\frac{10^{-4}}{\pi}F; R=100\Omega$$
$$V_{RMS}=220\ V; f=50\,Hz$$

$$X_L=2\pi fL=2\pi\times50\times\frac{2}{\pi}=200\Omega$$

$$X_C=\frac{1}{2\pi fC}=\frac{1}{2\pi\times50\times\frac{10^{-4}}{\pi}}=100\Omega$$

(i) Impedance, $Z=\sqrt{R^2+\left(X_L-X_C\right)^2}$

$$=\sqrt{100^2+\left(200-100\right)^2}\ =141.4\ \Omega$$

(ii) Peak value of current,

$$I_m=\frac{V_m}{Z}=\frac{\sqrt{2}V_{RMS}}{Z}$$
$$=\frac{\sqrt{2}\times220}{141.4}\ =2.2\ \text{A}$$

(iii) Power factor of the circuit,

$$\cos\phi=\frac{R}{Z}=\frac{100}{141.4}=0.707$$

(iv) Power factor at resonance

$$\cos\phi=\frac{R}{Z}=\frac{R}{R}=1$$

4.9 OSCILLATION IN LC CIRCUITS

4.9.1 Energy conversion during LC oscillations

We have learnt that energy can be stored in both inductors and capacitors (Refer sections 1.8.2 and 4.3.2). In inductors, the energy is stored in the form of magnetic field while in capacitors; it is stored as the electric field.

Whenever energy is given to a circuit containing a pure inductor of inductance *L* and a capacitor of capacitance *C*, the energy oscillates back and forth between the magnetic field of the inductor and the electric field of the capacitor. Thus the electrical oscillations of definite frequency are generated. These oscillations are called *LC* oscillations.

Generation of LC oscillations

Let us assume that the capacitor is fully charged with maximum charge Q_m at the initial stage. So that the energy stored in the capacitor is maximum and is given by $U_E=Q_m^2/2C$. As there is no current in the inductor, the energy stored in it is zero i.e., $U_B=0$. Therefore, the total energy is wholly electrical. This is shown in Figure 4.57(a).

The capacitor now begins to discharge through the inductor that establishes current *i* in clockwise direction. This current produces a magnetic field around the inductor and the energy stored in the inductor is given by $U_B=Li^2/2$. As the charge in the capacitor decreases, the energy stored in it also decreases and is given by

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